Royal University of Bhutan

Assignment 02

Module: Integrated system

Lab report 14

Module code: CTE 412

Std no: 02230136

1. Objective

The Student Services Office currently takes the Campus Notice Board offline during each release and has not measured the resulting downtime. This project deploys the Notice Board API on a three-node Kind cluster to compare four deployment approaches: Recreate, RollingUpdate, blue/green deployment, and feature flags. A monitoring service sends requests to the API twice per second to measure availability and observe how each release behaves.

For each deployment approach, the following are measured:

- Failed requests: Number of requests that fail during the monitored period.
- Mixed-version window: Duration during which responses from two application versions are observed.
- Peak Pod count: Maximum number of application Pods required during deployment, including terminating Pods.

The project also tests a deliberately broken release to demonstrate how Kubernetes can detect an unsuccessful rollout and roll back to a healthy version. The results are used to evaluate deployment reliability and recovery. Course coverage: Unit IV, Sections 4.1–4.3 (Rolling Updates, blue/green deployments, feature flags, monitoring, and containerised CI/CD); LO6 and LO7. Supporting evidence is available in the evidence/ directory.

2. Process

2.1 Cluster and quota

```
PS D:\4th year\Manoj\lab 14\cte412-notice-board> & "C:\Program Files\Git\bin\bash.exe" --login -i
RSION                          CONTAINER-RUNTIME
cte412-02230136-control-plane   Ready    control-plane   7m8s    v1.37.0   172.18.0.4    <none>        Debian GNU/Linux 13 (trixie)   6.18.33.2
-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230136-worker          Ready    <none>          6m54s   v1.37.0   172.18.0.2    <none>        Debian GNU/Linux 13 (trixie)   6.18.33.2
-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
cte412-02230136-worker2         Ready    <none>          6m54s   v1.37.0   172.18.0.3    <none>        Debian GNU/Linux 13 (trixie)   6.18.33.2
-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
```

```
$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS         PORTS
       NAMES
9b3349a9a9e4   kindest/node:v1.37.0   "/usr/local/bin/entr…"   8 minutes ago   Up 8 minutes   127.0.0.1:30090->30090/tcp, 127.0.0.1:52468->6443
/tcp   cte412-02230136-control-plane
9733718a9a77   kindest/node:v1.37.0   "/usr/local/bin/entr…"   8 minutes ago   Up 8 minutes
       cte412-02230136-worker
1bb72a3e1699   kindest/node:v1.37.0   "/usr/local/bin/entr…"   8 minutes ago   Up 8 minutes
       cte412-02230136-worker2
```

Figure 1. Three kind nodes and `docker ps` showing `127.0.0.1:30090->30090/tcp`. Evidence: t1-cluster.txt

```
$ kubectl describe resourcequota nb-quota
Name:            nb-quota
Namespace:       nb-02230136
Resource         Used  Hard
--------         ----  ----
count/services   0     4
limits.cpu       0     2
limits.memory    0     2Gi
pods             0     15
requests.cpu     0     1
requests.memory  0     1Gi
```

Figure 2. Resource quota `nb-quota`: 15 Pods, 4 Services, 1 CPU / 1Gi requests, 2 CPU / 2Gi limits. Evidence: t1-quota.txt

2.2 Image checks

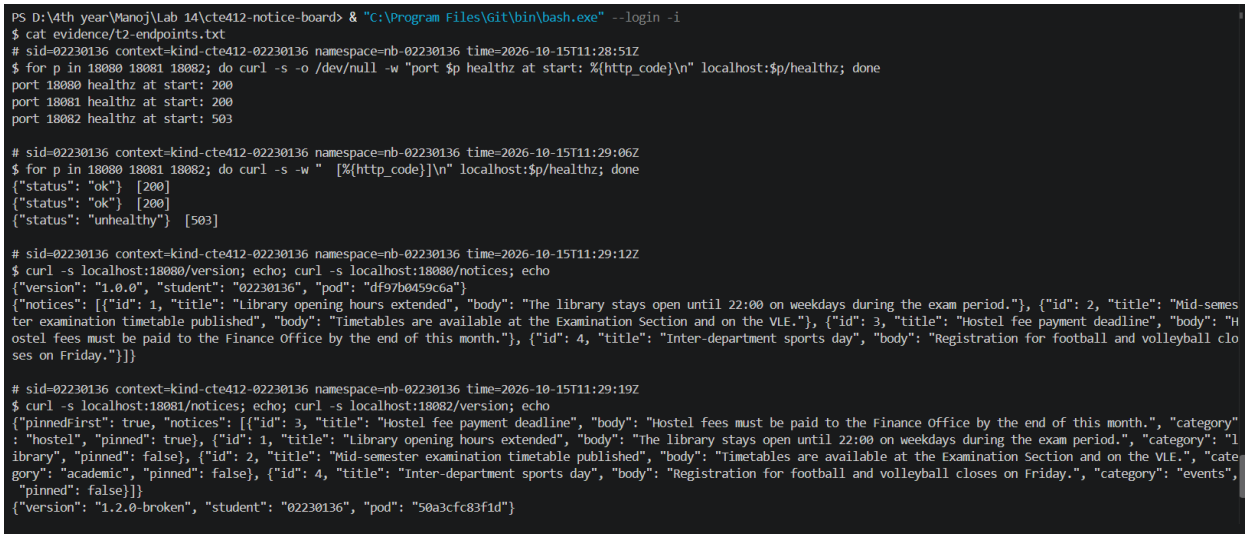


Figure 3. Local test of the three images: /healthz (200, 200, 503 for the broken one) and /notices for 1.1.0 with "pinnedFirst": true. Evidence: t2-endpoints.txt

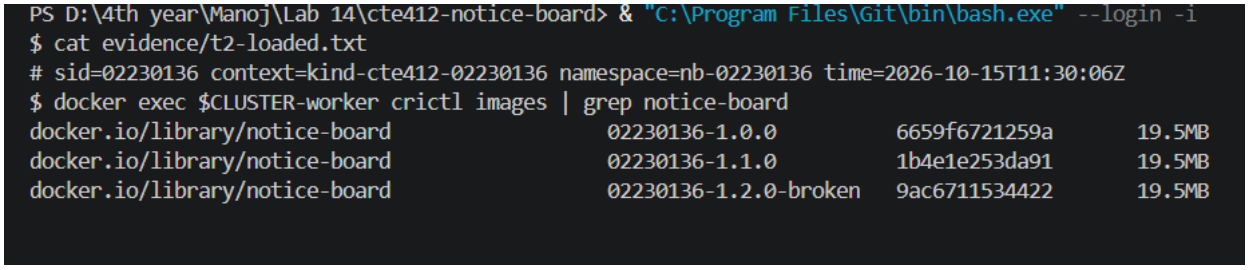


Figure 4. crictl images on a worker node: the three images, loaded locally (no registry). Evidence: t2-loaded.txt

2.3 Baseline deployment

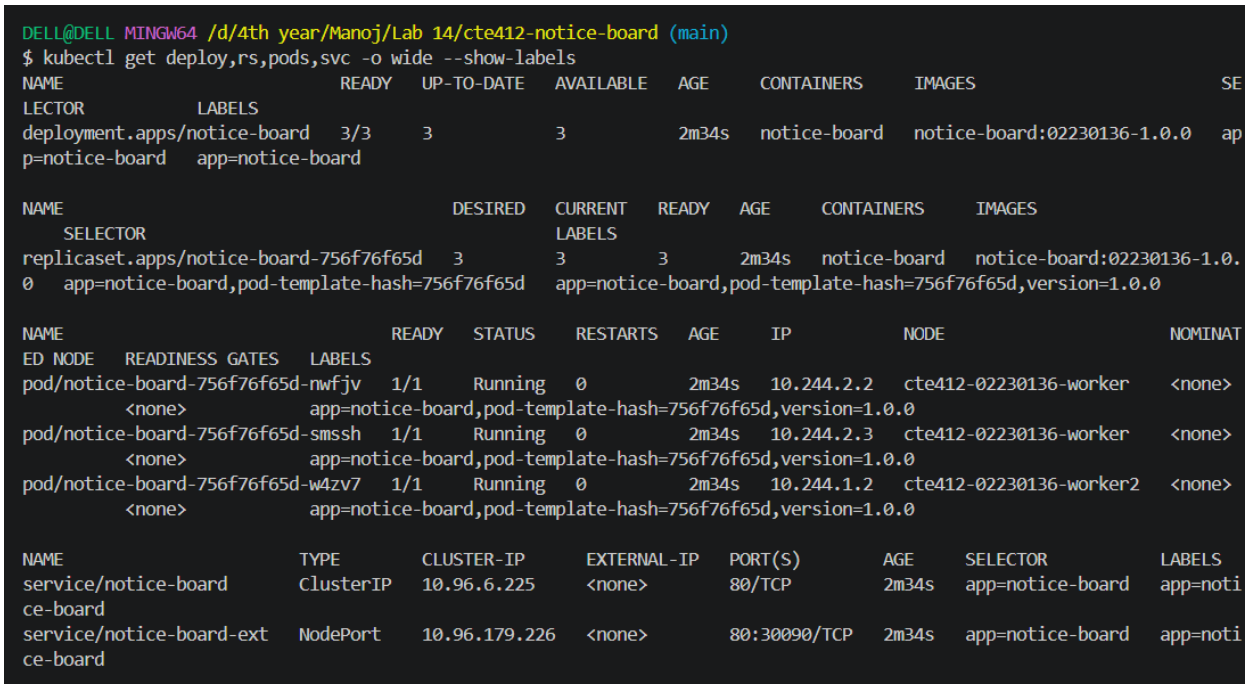


Figure 5. kubectl get deploy,rs,pods,svc -o wide after Task 3. Evidence: t3-deploy.txt

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ for i in $(seq 1 30); do
  curl -s http://localhost:30090/version
  echo
done | grep -o '"pod": "[^"]*"' | sort | uniq -c
     11 "pod": "notice-board-756f76f65d-nwfjv"
      7 "pod": "notice-board-756f76f65d-smssh"
     12 "pod": "notice-board-756f76f65d-w4zv7"
```

Figure 6. curl loop on localhost:30090/version, answered by different Pods. Evidence: t3-host-version.txt

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ kubectl logs monitor --tail=5
2026-10-15T12:55:02Z OK {"version": "1.0.0", "student": "02230136", "pod": "notice-board-756f76f65d-smssh"}
2026-10-15T12:55:02Z OK {"version": "1.0.0", "student": "02230136", "pod": "notice-board-756f76f65d-nwfjv"}
2026-10-15T12:55:03Z OK {"version": "1.0.0", "student": "02230136", "pod": "notice-board-756f76f65d-smssh"}
2026-10-15T12:55:03Z OK {"version": "1.0.0", "student": "02230136", "pod": "notice-board-756f76f65d-nwfjv"}
2026-10-15T12:55:04Z OK {"version": "1.0.0", "student": "02230136", "pod": "notice-board-756f76f65d-w4zv7"}
```

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ cat evidence/t4-rest-summary.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T12:54:55Z
$ scripts/summarise.sh evidence/t4-rest-monitor.txt
Total requests       : 107
Failed requests      : 0
First timestamp      : 2026-10-15T12:53:42Z
Last timestamp       : 2026-10-15T12:54:44Z
Duration             : 62 s
First / last FAIL    : none
Responses per version:
  1.0.0              107   first 2026-10-15T12:53:42Z   last 2026-10-15T12:54:44Z
Mixed-version window : 0 s (only one version observed)
```

Figure 7. Monitor output at rest and its summary (requests, 0 failed). Evidence: t4-rest-monitor.txt, t4-rest-summary.txt

3. Results

Rows come from scripts/summarise.sh on a monitor log that starts at least 10 s before the apply and ends at least 10 s after the rollout. Peak Pods = highest count of app=notice-board Pods (Terminating included) in the matching -pods.txt.

Timestamps have 1 s resolution (±1 s). In the flag row both sides report 1.1.0, so the mixed window is between flag-off and flag-on Pods (from t8-flag-pods.txt and rollout duration).

3.1 Recreate and RollingUpdate

```
$ cat evidence/t5-recreate-summary.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T13:17:22Z
$ scripts/summarise.sh evidence/t5-recreate-monitor.txt
Total requests       : 101
Failed requests      : 7
First timestamp      : 2026-10-15T13:16:21Z
Last timestamp       : 2026-10-15T13:17:22Z
Duration             : 61 s
First / last FAIL    : 2026-10-15T13:16:49Z / 2026-10-15T13:16:58Z (9 s apart)
Responses per version:
  1.0.0              53   first 2026-10-15T13:16:21Z   last 2026-10-15T13:16:48Z
  1.1.0              41   first 2026-10-15T13:17:00Z   last 2026-10-15T13:17:22Z
Mixed-version window : 0 s (1.0.0 and 1.1.0 both observed)
```

Figure 8. Recreate: failures and a gap with no version. Evidence: t5-recreate-summary.txt

```
$ cat evidence/t5-rolling-summary.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T13:21:49Z
$ scripts/summarise.sh evidence/t5-rolling-monitor.txt
Total requests       : 163
Failed requests      : 0
First timestamp      : 2026-10-15T13:20:25Z
Last timestamp       : 2026-10-15T13:21:49Z
Duration             : 84 s
First / last FAIL    : none
Responses per version:
  1.0.0               87   first 2026-10-15T13:20:25Z   last 2026-10-15T13:21:27Z
  1.1.0               76   first 2026-10-15T13:20:56Z   last 2026-10-15T13:21:49Z
Mixed-version window : 31 s (1.0.0 and 1.1.0 both observed)
```

Figure 9. RollingUpdate: 0 failures, both versions overlapping. Evidence: t5-rolling-summary.txt

3.2 Broken release and rollback

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ kubectl rollout history deployment/notice-board -n nb-02230136
deployment.apps/notice-board
REVISION  CHANGE-CAUSE
3         task 5.2: back to 1.0.0
4         task 5.3: release 1.1.0 with RollingUpdate 0/1, minReady 5
```

Figure 10. kubectl rollout history with change-causes. Evidence: t5-history.txt, t6-history.txt

```
$ cat evidence/t6-rollout-status.txt
cat evidence/t6-condition.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T13:32:17Z
$ kubectl rollout status deployment/notice-board --timeout=120s
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
error: deployment "notice-board" exceeded its progress deadline

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T13:34:20Z
$ kubectl rollout status deployment/notice-board --timeout=120s
error: deployment "notice-board" exceeded its progress deadline

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T13:38:32Z
$ kubectl get deploy notice-board -n nb-02230136 -o jsonpath='{range .status.conditions[*]}{.type}{"\t"}{.status}{"\t"}
{.reason}{"\t"}{.message}{"\n"}{end}'
Available       True    MinimumReplicasAvailable        Deployment has minimum availability.
Progressing     False   ProgressDeadlineExceeded        ReplicaSet "notice-board-698c89d644" has timed out progressing.
```

Figure 11. kubectl rollout status ends in “exceeded its progress deadline”; condition Progressing False ProgressDeadlineExceeded. Evidence: t6-rollout-status.txt, t6-condition.txt

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ cat evidence/t6-pods.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T13:37:17Z
$ kubectl get pods -n nb-02230136 -o wide
NAME                            READY   STATUS    RESTARTS   AGE    IP           NODE                       NOMINATED NODE   READINESS GATES
monitor                         1/1     Running   0          44m    10.244.2.4   cte412-02230136-worker     <none>           <none>
notice-board-596979fb4d-m4bhr   1/1     Running   0          16m    10.244.1.6   cte412-02230136-worker2    <none>           <none>
notice-board-596979fb4d-nnzsz   1/1     Running   0          16m    10.244.2.9   cte412-02230136-worker     <none>           <none>
notice-board-596979fb4d-v6pm7   1/1     Running   0          16m    10.244.2.8   cte412-02230136-worker     <none>           <none>
notice-board-698c89d644-z78gd   0/1     Running   0          5m5s   10.244.1.7   cte412-02230136-worker2    <none>           <none>
```

Figure 12. Stalled rollout: three 1.1.0 Pods 1/1, one 1.2.0-broken Pod 0/1 Running. Evidence: t6-pods.txt

```
$ cat evidence/t6-diff.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T13:44:02Z
$ kubectl diff -f manifests/rolling/01-deployment.yaml; echo "kubectl diff exit code: $?"
diff -u -N C:\Users\DELL\AppData\Local\Temp\LIVE-1143696733/apps.v1.Deployment.nb-02230136.notice-board C:\Users\DELL\AppData\Local\Temp\MERGED-900457583/apps.v1
.Deployment.nb-02230136.notice-board
--- C:\Users\DELL\AppData\Local\Temp\LIVE-1143696733/apps.v1.Deployment.nb-02230136.notice-board        2026-10-15 19:44:02.321442500 +0600
+++ C:\Users\DELL\AppData\Local\Temp\MERGED-900457583/apps.v1.Deployment.nb-02230136.notice-board       2026-10-15 19:44:02.321973200 +0600
@@ -5,10 +5,9 @@
     deployment.kubernetes.io/revision: "6"
     kubectl.kubernetes.io/last-applied-configuration: |
       {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{"kubernetes.io/change-cause":"task 6.1: release 1.2.0-broken"},"labels":{"app":"not
ice-board"},"name":"notice-board","namespace":"nb-02230136"},"spec":{"minReadySeconds":5,"progressDeadlineSeconds":60,"replicas":3,"selector":{"matchLabels":{"ap
p":"notice-board"}},"strategy":{"rollingUpdate":{"maxSurge":1,"maxUnavailable":0},"type":"RollingUpdate"},"template":{"metadata":{"labels":{"app":"notice-board",
"version":"1.2.0-broken"}},"spec":{"containers":[{"image":"notice-board:02230136-1.2.0-broken","imagePullPolicy":"Never","livenessProbe":{"failureThreshold":3,"i
nitialDelaySeconds":10,"periodSeconds":10,"tcpSocket":{"port":"http"}},"name":"notice-board","ports":[{"containerPort":8080,"name":"http"}],"readinessProbe":{"fa
ilureThreshold":2,"httpGet":{"path":"/healthz","port":"http"},"periodSeconds":2,"timeoutSeconds":1},"resources":{"limits":{"cpu":"150m","memory":"128Mi"},"reques
ts":{"cpu":"50m","memory":"64Mi"}},"securityContext":{"allowPrivilegeEscalation":false,"runAsNonRoot":true}}]}}}}
-    kubernetes.io/change-cause: 'task 5.3: release 1.1.0 with RollingUpdate 0/1, minReady
-      5'
+    kubernetes.io/change-cause: PASTE_CHANGE_CAUSE_HERE
   creationTimestamp: "2026-10-15T11:37:23Z"
-  generation: 6
+  generation: 7
   labels:
     app: notice-board
   name: notice-board
kubectl diff exit code: 1
```

Figure 13. kubectl diff -f manifests/rolling/01-deployment.yaml and its exit code. Evidence: t6-diff.txt

3.3 Blue/green

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ kubectl get deploy,svc,pods -o wide -L slot,version -n nb-02230136
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS     IMAGES                          SELECTOR                          SLOT    VE
RSION
deployment.apps/notice-board-blue   3/3     3            3           11m   notice-board   notice-board:02230136-1.0.0     app=notice-board,slot=blue        blue
deployment.apps/notice-board-green  3/3     3            3           11m   notice-board   notice-board:02230136-1.1.0     app=notice-board,slot=green       green

NAME                           TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE   SELECTOR                          SLOT   VERSION
service/notice-board           ClusterIP   10.96.6.225     <none>        80/TCP         169m  app=notice-board,slot=green
service/notice-board-ext       NodePort    10.96.179.226   <none>        80:30090/TCP   169m  app=notice-board,slot=green
service/notice-board-preview   ClusterIP   10.96.246.130   <none>        80/TCP         11m   app=notice-board,slot=green

NAME                                 READY   STATUS    RESTARTS   AGE   IP            NODE                       NOMINATED NODE   READINESS GATES   SLOT    VE
RSION
pod/monitor                          1/1     Running   0          93m   10.244.2.4    cte412-02230136-worker     <none>           <none>
pod/notice-board-blue-5fcf54b448-b6bfh   1/1     Running   0          11m   10.244.1.8    cte412-02230136-worker2    <none>           <none>            blue    1.
0.0
pod/notice-board-blue-5fcf54b448-m7jwj   1/1     Running   0          11m   10.244.1.10   cte412-02230136-worker2    <none>           <none>            blue    1.
0.0
pod/notice-board-blue-5fcf54b448-vllzx   1/1     Running   0          11m   10.244.2.12   cte412-02230136-worker     <none>           <none>            blue    1.
0.0
pod/notice-board-green-b49f6dcc6-89vcj   1/1     Running   0          11m   10.244.2.11   cte412-02230136-worker     <none>           <none>            green   1.
1.0
pod/notice-board-green-b49f6dcc6-mll5j   1/1     Running   0          11m   10.244.2.10   cte412-02230136-worker     <none>           <none>            green   1.
1.0
pod/notice-board-green-b49f6dcc6-qj8l6   1/1     Running   0          11m   10.244.1.9    cte412-02230136-worker2    <none>           <none>            green   1.
1.0

DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
```

Figure 14. kubectl get deploy,svc,pods -L slot,version with blue and green running. Evidence: t7-objects.txt

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ cat evidence/t7-preview.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:18:39Z
$ for i in 1 2 3; do kubectl exec -n nb-02230136 monitor -- wget -qO- http://notice-board-preview/version; echo; done
{"version": "1.1.0", "student": "02230136", "pod": "notice-board-green-b49f6dcc6-89vcj"}
{"version": "1.1.0", "student": "02230136", "pod": "notice-board-green-b49f6dcc6-qj8l6"}
{"version": "1.1.0", "student": "02230136", "pod": "notice-board-green-b49f6dcc6-qj8l6"}

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:18:47Z
$ for i in 1 2 3; do kubectl exec -n nb-02230136 monitor -- wget -qO- http://notice-board/version; echo; done
{"version": "1.0.0", "student": "02230136", "pod": "notice-board-blue-5fcf54b448-m7jwj"}
{"version": "1.0.0", "student": "02230136", "pod": "notice-board-blue-5fcf54b448-b6bfh"}
{"version": "1.0.0", "student": "02230136", "pod": "notice-board-blue-5fcf54b448-m7jwj"}

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:18:56Z
$ kubectl exec -n nb-02230136 monitor -- wget -qO- http://notice-board-preview/notices; echo
{"pinnedFirst": false, "notices": [{"id": 1, "title": "Library opening hours extended", "body": "The library stays open until 22:00 on weekdays during the exam p
eriod.", "category": "library", "pinned": false}, {"id": 2, "title": "Mid-semester examination timetable published", "body": "Timetables are available at the Exa
mination Section and on the VLE.", "category": "academic", "pinned": false}, {"id": 3, "title": "Hostel fee payment deadline", "body": "Hostel fees must be paid
to the Finance Office by the end of this month.", "category": "hostel", "pinned": true}, {"id": 4, "title": "Inter-department sports day", "body": "Registration
for football and volleyball closes on Friday.", "category": "events", "pinned": false}]}
```

Figure 15. Preview Service answers 1.1.0 while production answers 1.0.0. Evidence: t7-preview.txt

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ cat evidence/t7-switches-all-summary.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:25:00Z
$ scripts/summarise.sh evidence/t7-switches-all-monitor.txt
Total requests       : 669
Failed requests      : 0
First timestamp      : 2026-10-15T14:18:58Z
Last timestamp       : 2026-10-15T14:24:54Z
Duration             : 356 s
First / last FAIL    : none
Responses per version:
  1.0.0              114   first 2026-10-15T14:18:58Z   last 2026-10-15T14:20:24Z
  1.1.0              555   first 2026-10-15T14:19:32Z   last 2026-10-15T14:24:54Z
Mixed-version window : 52 s (1.0.0 and 1.1.0 both observed)
```

Figure 16. Monitor summary across all three switches. Evidence: t7-switches-all-summary.txt

3.4 Feature flag

```
$ cat evidence/t8-after-configmap.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:34:21Z
$ kubectl get configmap notice-board-flags -o jsonpath="{.data}"; echo
{"FEATURE_PINNED":"true"}

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:34:25Z
$ for p in $(kubectl get pods -l slot=green -o name); do echo "$p FEATURE_PINNED=$(kubectl exec ${p#pod/} -- printenv F
EATURE_PINNED)"; done
pod/notice-board-green-846d6cbfb9-2lcc2 FEATURE_PINNED=false
pod/notice-board-green-846d6cbfb9-qn2l6 FEATURE_PINNED=false
pod/notice-board-green-846d6cbfb9-stgwr FEATURE_PINNED=false

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:34:30Z
$ kubectl get pods -l slot=green
NAME                                  READY   STATUS    RESTARTS   AGE
notice-board-green-846d6cbfb9-2lcc2   1/1     Running   0          4m15s
notice-board-green-846d6cbfb9-qn2l6   1/1     Running   0          4m29s
notice-board-green-846d6cbfb9-stgwr   1/1     Running   0          4m2s

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:34:36Z
$ curl -s http://localhost:30090/notices; echo
{"pinnedFirst": false, "notices": [{"id": 1, "title": "Library opening hours extended", "body": "The library stays open
 until 22:00 on weekdays during the exam period.", "category": "library", "pinned": false}, {"id": 2, "title": "Mid-sem
ester examination timetable published", "body": "Timetables are available at the Examination Section and on the VLE.",
"category": "academic", "pinned": false}, {"id": 3, "title": "Hostel fee payment deadline", "body": "Hostel fees must b
e paid to the Finance Office by the end of this month.", "category": "hostel", "pinned": true}, {"id": 4, "title": "Int
er-department sports day", "body": "Registration for football and volleyball closes on Friday.", "category": "events",
"pinned": false}]}
```

Figure 17. ConfigMap shows FEATURE_PINNED: true, but running green Pods still print false (same names and ages). Evidence: t8-after-configmap.txt

```
$ cat evidence/t8-after-rollout.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:40:00Z
$ for p in $(kubectl get pods -l slot=green -o name); do echo "$p FEATURE_PINNED=$(kubectl exec ${p#pod/} -- printenv F
EATURE_PINNED)"; done
pod/notice-board-green-6cd6767674-knzn5 FEATURE_PINNED=true
pod/notice-board-green-6cd6767674-vznlx FEATURE_PINNED=true
pod/notice-board-green-6cd6767674-xr54m FEATURE_PINNED=true

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:40:04Z
$ curl -s http://localhost:30090/notices; echo
{"pinnedFirst": true, "notices": [{"id": 3, "title": "Hostel fee payment deadline", "body": "Hostel fees must be paid t
o the Finance Office by the end of this month.", "category": "hostel", "pinned": true}, {"id": 1, "title": "Library ope
ning hours extended", "body": "The library stays open until 22:00 on weekdays during the exam period.", "category": "li
brary", "pinned": false}, {"id": 2, "title": "Mid-semester examination timetable published", "body": "Timetables are av
ailable at the Examination Section and on the VLE.", "category": "academic", "pinned": false}, {"id": 4, "title": "Inte
r-department sports day", "body": "Registration for football and volleyball closes on Friday.", "category": "events", "
pinned": false}]}


DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
```

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ cat evidence/t8-flag-summary.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:39:10Z
$ scripts/summarise.sh evidence/t8-flag-monitor.txt
Total requests       : 470
Failed requests      : 0
First timestamp      : 2026-10-15T14:35:04Z
Last timestamp       : 2026-10-15T14:39:09Z
Duration             : 245 s
First / last FAIL    : none
Responses per version:
  1.1.0              470   first 2026-10-15T14:35:04Z   last 2026-10-15T14:39:09Z
Mixed-version window : 0 s (only one version observed)
```

Figure 18. After the flags-revision rollout: Pods print true, /notices returns "pinnedFirst": true with notice 3 first, 0 failed requests. Evidence: t8-after-rollout.txt, t8-flag-summary.txt

3.5 Rebuild and clean-up

```
$ cat evidence/t9-rebuilt.txt
# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:50:59Z
$ kubectl get deploy,svc,pods -o wide -L slot,version
NAME                                    READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS     IMAGES                                SELECTOR                          SLOT    VERSION
deployment.apps/notice-board-blue       3/3     3            3           28s   notice-board   notice-board:02230136-1.0.0   app=notice-board,slot=blue    blue
deployment.apps/notice-board-green      3/3     3            3           28s   notice-board   notice-board:02230136-1.1.0   app=notice-board,slot=green   green

NAME                           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE   SELECTOR                          SLOT   VERSION
service/notice-board           ClusterIP   10.96.178.30   <none>        80/TCP         28s   app=notice-board,slot=green
service/notice-board-ext       NodePort    10.96.67.172   <none>        80:30090/TCP   28s   app=notice-board,slot=green
service/notice-board-preview   ClusterIP   10.96.88.104   <none>        80/TCP         28s   app=notice-board,slot=green

NAME                                    READY   STATUS             RESTARTS   AGE   IP            NODE                       NOMINATED NODE   READINESS GATES   SLOT    VERSION
pod/monitor                             0/1     ImagePullBackOff   0          32s   10.244.1.15   cte412-02230136-worker2    <none>           <none>
pod/notice-board-blue-5fcf54b448-8nmqf  1/1     Running            0          28s   10.244.2.15   cte412-02230136-worker     <none>           <none>            blue    1.0.0
pod/notice-board-blue-5fcf54b448-lw8hm  1/1     Running            0          28s   10.244.2.16   cte412-02230136-worker     <none>           <none>            blue    1.0.0
pod/notice-board-blue-5fcf54b448-wjmhb  1/1     Running            0          28s   10.244.1.16   cte412-02230136-worker2    <none>           <none>            blue    1.0.0
pod/notice-board-green-6cd6767674-q6g6m 1/1     Running            0          28s   10.244.1.17   cte412-02230136-worker2    <none>           <none>            green   1.1.0
pod/notice-board-green-6cd6767674-ttmh6 1/1     Running            0          28s   10.244.2.17   cte412-02230136-worker     <none>           <none>            green   1.1.0
pod/notice-board-green-6cd6767674-x6gf9 1/1     Running            0          28s   10.244.2.18   cte412-02230136-worker     <none>           <none>            green   1.1.0

# sid=02230136 context=kind-cte412-02230136 namespace=nb-02230136 time=2026-10-15T14:51:04Z
$ curl -s http://localhost:30090/version; echo; curl -s http://localhost:30090/notices; echo
{"version": "1.1.0", "student": "02230136", "pod": "notice-board-green-6cd6767674-q6g6m"}
{"pinnedFirst": true, "notices": [{"id": 3, "title": "Hostel fee payment deadline", "body": "Hostel fees must be paid to the Finance Office by the end of this month.", "category": "hostel", "pinned": t
rue}, {"id": 1, "title": "Library opening hours extended", "body": "The library stays open until 22:00 on weekdays during the exam period.", "category": "library", "pinned": false}, {"id": 2, "title":
"Mid-semester examination timetable published", "body": "Timetables are available at the Examination Section and on the VLE.", "category": "academic", "pinned": false}, {"id": 4, "title": "Inter-depart
ment sports day", "body": "Registration for football and volleyball closes on Friday.", "category": "events", "pinned": false}]}
```

Figure 19. State rebuilt from the README commands only. Evidence: t9-emptied.txt, t9-rebuilt.txt

```
DELL@DELL MINGW64 /d/4th year/Manoj/Lab 14/cte412-notice-board (main)
$ kind get clusters
No kind clusters found.
```

Figure 20. kind get clusters → "No kind clusters found." Evidence: t9-deleted.txt

4. Conclusion

The results show that blue/green deployment and RollingUpdate can reduce downtime compared with Recreate. Recreate temporarily removes the old version before the new one is ready, causing requests to fail. RollingUpdate maintains availability by starting new Pods before removing old ones, although both versions may serve traffic during the rollout. Blue/green keeps both versions running and switches production traffic by changing the Service selector, allowing faster switching and rollback at the cost of additional resources.

The broken-release test demonstrated the importance of readiness probes and rollout monitoring. The unready version received no traffic, while the healthy Pods continued serving requests until the rollout was rolled back. The feature-flag test also showed that changing a ConfigMap does not automatically update environment variables in running containers; a Pod restart or rollout is required. Overall, the experiment shows that deployment strategies should be selected according to application availability, compatibility, resource limits, and recovery requirements. Blue/green is useful when rapid switching is important, RollingUpdate offers a balance between availability and resource usage, and Recreate is suitable when two versions must not run simultaneously.